The documentation and process conversion measures necessary to comply with this document shall be completed by 16 November 2023.

INCH-POUND

MIL-PRF-19500/408L
w/AMENDMENT 1
16 August 2023
SUPERSEDING
MIL-PRF-19500/408L
18 March 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, HIGH-POWER,
TYPES 2N3715 AND 2N3716, CASE MOUNT THROUGH HOLE PACKAGE,
QUALITY LEVELS JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, high-power transistors. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each device type as specified in MIL-PRF-19500.

1.2 Package outline. The device package outline is a TO-204AA (formerly TO-3) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

Types	P_T (1) T_A = +25°C	P_T (2) T_C = +25°C	$R_{\theta JC}$ (3)	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_J and T_{STG}
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	°C
2N3715	5.0	117	1.5	80	60	7.0	4.0	10	-65 to +200
2N3716	5.0	117	1.5	100	80	7.0	4.0	10	-65 to +200

(1) Derate linearly 28.57 mW/°C above T_A = +25°C.
(2) See figure 2 for temperature-power derating curves.
(3) See figure 3 for transient thermal impedance graph.

1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Limit	h_{FE2} (1) V_{CE} = 2.0 V dc I_C = 3.0 A dc	h_{FE4} (1) V_{CE} = 4.0 V dc I_C = 10 A dc	$V_{BE(SAT)1}$ (1) I_C = 5.0 A dc I_B = 0.5 A dc	$V_{CE(SAT)1}$ (1) I_C = 5.0 A dc I_B = 0.5 A dc	C_{obo} V_{CB} = 10 V dc, I_E = 0 f = 1 MHz	$\|h_{fe}\|$ V_{CE} = 10 V dc I_C = 0.5 A dc f = 1 MHz
			V dc	V dc	pF	
Min	30	5.0				4.0
Max	120		1.5	1.0	500	20

(1) Pulsed (see 4.5.1).

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example, 6.5 for a list of available PINs and 6.6 for interchangeability information.

1.5.1 JAN certification mark and quality level. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "3715" and "3716".

1.5.3 Suffix symbols. Suffix symbols are not applicable to this specification sheet.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

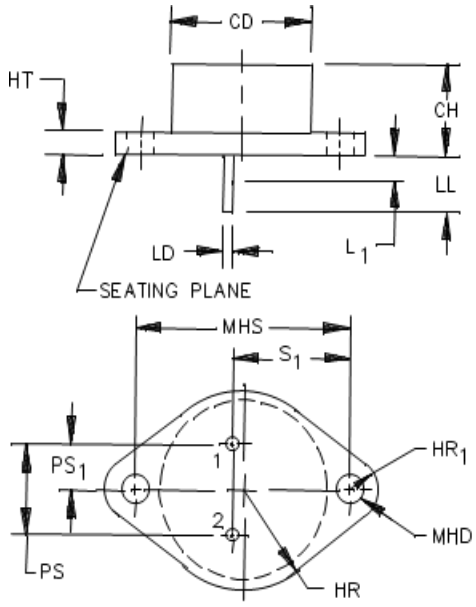
MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Notes
	in	in	mm	mm	
	Min	Max	Min	Max	
CD		.875		22.23	3
CH	.250	.450	6.35	11.43	
HR	.495	.525	12.57	13.34	
HR_1	.131	.188	3.33	4.78	4
HT	.060	.135	1.52	3.43	
L_1		.050		1.27	5, 6
LD	.038	.043	0.97	1.09	5, 6
LL	.312	.500	7.92	12.70	5
MHD	.151	.165	3.84	4.19	4
MHS	1.177	1.197	29.90	30.40	
PS	.420	.440	10.67	11.18	7
PS_1	.205	.225	5.21	5.72	7
S_1	.655	.675	16.64	17.15	7



NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Terminal 1 is the emitter; terminal 2 is the base; and the collector shall be electrically connected to the case.
3. Body contour is optional within zone defined by dimension CD.
4. Applies to both ends.
5. Applies to both terminals.
6. Dimension LD applies between dimension L_1 and LL. Lead diameter shall not exceed twice dimension LD within dimension L_1. Diameter is uncontrolled in dimension L_1.
7. These dimensions shall be measured at points .050 inch (1.27 mm) to .055 inch (1.4 mm) below the seating plane. When gauge is not used, measurement will be made at the seating plane.
8. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.

FIGURE 1. Physical dimensions TO-204AA (formerly TO-3).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3)

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO-204AA, formerly TO-3) herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Pin out. The identification of terminals of the device package shall be as shown on figure 1. Terminal 1 shall be connected to the emitter, terminal 2 shall be connected to the base, and the collector shall be electrically connected to the case.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (quality levels JANS, JANTX and JANTXV only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANS level	Measurements for JANTX and JANTXV levels
3c (1)	Thermal impedance (see 4.3.1).	Thermal impedance (see 4.3.1).
9	I_{CES1} and h_{FE2}	I_{CES1}
11	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 1 µA dc, whichever is greater. Δh_{FE2} = ±15 percent of initial value.	I_{CES1} and h_{FE2}; ΔI_{CES1} = 100 percent of initial value or 1 µA dc, whichever is greater.
12	See 4.3.2	See 4.3.2
13	ΔI_{CES1} = 100 percent of initial value or 1 µA dc, whichever is greater; Δh_{FE2} = ±15 percent of initial value; subgroup 2 of table I herein	ΔI_{CES1} = 100 percent of initial value or 1 µA dc, whichever is greater; Δh_{FE2} = ±20 percent of initial value; subgroup 2 of table I herein

(1) Thermal impedance shall be performed anytime after temperature cycling (screen 3a), JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with test method 3131 of MIL-STD-750 using the guidelines in that test method for determining I_H, I_M, t_H, t_{SW}, (and V_H where appropriate). See table II, subgroup 4 (group E) herein.

4.3.2 Power burn-in conditions. Power burn-in conditions shall be as follows: T_J = +187.5 ±12.5°C; V_{CE} = 25 ±5 V dc; T_A = room ambient as defined in the general requirements of MIL-STD-750. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for quality levels JANTX and JANTXV. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. End-point electrical measurements shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX and JANTXV) of MIL-PRF-19500, and as follows. Delta measurements shall be in accordance with the applicable steps of 4.6 herein.

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4.4.2.1 Quality level JANS (table E-VIA of MIL-PRF-19500).

Subgroup	Method	Conditions
B4	1037	V_{CB} = 30 V dc, P_T = 4 W at T_A = +25°C ±3°C, t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air on the device shall be permitted. A separate sample may be selected for each test.
B5	1027	V_{CB} = 30 V dc, T_A = +125°C ±25°C for 96 hours; P_T = 4 W at T_A = +100°C or adjusted as required by the chosen T_A to give an average lot T_J = +225°C. Marking legibility requirement shall not apply.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV (table E-VIB of MIL-PRF-19500).

Subgroup	Method	Conditions
B3	1037	V_{CB} ≥ 10 V dc; ΔT_J between cycles ≥ 100°C; t_{on} = t_{off} ≥ 1 minute for 2,000 cycles. No heat sink or forced air cooling on the device shall be permitted.
B6	1032	T_A = +200°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows herein. Delta measurements shall be in accordance with the applicable step of 4.6 herein.

Subgroup	Method	Conditions
C2	1056	Test condition B.
C2	2036	Test condition A, weight = 10 lbs (4.54 Kg), application time = 15 seconds.
C5	3131	See 4.3.1, $R_{\theta JC}$ = 1.5°C/W.
C6	1037	V_{CB} ≥ 10 V dc; ΔT_J between cycles ≥ 100°C, t_{on} = t_{off} ≥ 1 minute for 6,000 cycles. No heat sink or forced air cooling on the device shall be permitted.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Delta measurements shall be in accordance with the applicable step of 4.6 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse response measurements. The conditions for pulse response measurement shall be as specified in section 4 of MIL-STD-750.

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4.6 Delta measurements. The delta measurements for groups B, C, and E shall be as specified below.

Step	Inspection (1) (2) (3) (4)	MIL-STD-750		Symbol	Limits	Limits	Unit
		Method	Conditions		Min	Max	
1	Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	Δh_{FE2} (5)	±25 percent change from initial value		
2	Saturation voltage, collector to emitter	3071	I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1)	$\Delta V_{CE(SAT)1}$ (5)	±50 percent change from initial value		

(1) The delta measurements for group B, quality level JANS, shall be as follows:
   a. In addition to the measurements specified for subgroup 4 of table E-VIA of MIL-PRF-19500, the measurements of step 1 shall also be taken.
   b. In addition to the measurements specified for subgroup 5 of table E-VIA of MIL-PRF-19500, the measurements of steps 1 and 2 shall also be taken.

(2) The delta measurements for group B, quality levels JAN, JANTX and JANTXV shall be as follows:
   a. In addition to the measurements specified for subgroup 3 of table E-VIB of MIL-PRF-19500, the measurements of steps 1 shall also be taken.
   b. In addition to the measurements specified for subgroup 6 of table E-VIB of MIL-PRF-19500, the measurements of step 1 shall also be taken.

(3) The delta measurements for group C shall be as follows:
   a. In addition to the measurements specified for subgroup 6 of table E-VII of MIL-PRF-19500, the measurements of all steps shall also be taken for JANS.
   b. In addition to the measurements specified for subgroup 6 of table E-VII of MIL-PRF-19500, the measurements of step 1 shall also be taken for JAN, JANTX, and JANTXV.

(4) The delta measurements for group E, all quality levels, shall be as follows: In addition to the measurements specified for subgroups 1 and 2 of table E-IX of MIL-PRF-19500, the measurements of steps 1 and 2 shall also be taken.

(5) Devices which exceed the table I (group A) limits for this test shall not be acceptable.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3131	See 4.3.1	$Z_{\theta JX}$			°C/W
Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3715				60		V dc
2N3716				80		V dc
Collector-base cutoff current	3036	Bias condition D	I_{CBO}			
2N3715		V_{CB} = 80 V dc			10	µA dc
2N3716		V_{CB} = 100 V dc			10	µA dc
Emitter-base cutoff current	3061	Bias condition D; V_{EB} = 7 V dc	I_{EBO}		1.0	mA dc
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = 1.5 V dc	I_{CEX1}			
2N3715		V_{CE} = 60 V dc			10	µA dc
2N3716		V_{CE} = 80 V dc			10	µA dc
Collector to emitter cutoff current	3041	Bias condition C	I_{CES1}			
2N3715		V_{CE} = 60 V dc			10	µA dc
2N3716		V_{CE} = 80 V dc			10	µA dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 5 A dc; I_B = 0.5 A dc; pulsed (see 4.5.1);	$V_{BE(SAT)1}$		1.5	V dc
Base emitter voltage (saturated)	3066	Test condition A; I_C = 10 A dc; I_B = 2 A dc; pulsed (see 4.5.1);	$V_{BE(SAT)2}$		3.0	V dc
Saturation voltage; collector to emitter	3071	I_C = 5 A dc; I_B = 0.5 A dc pulsed (see 4.5.1)	$V_{CE(SAT)1}$		1.0	V dc
Saturation voltage; collector to emitter	3071	I_C = 10A dc; I_B = 2 A dc pulsed (see 4.5.1)	$V_{CE(SAT)2}$		2.5	V dc
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE1}	50	150	
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0A dc; pulsed (see 4.5.1)	h_{FE2}	30	120	

See footnote at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued						
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 5 A dc; pulsed (see 4.5.1)	h_{FE3}	10		
Forward-current transfer ratio	3076	V_{CE} = 4.0 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE4}	5		
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition C	I_{CES2}			
2N3715		V_{CE} = 50 V dc			5.0	mA dc
2N3716		V_{CE} = 70 V dc			5.0	mA dc
Low temperature operation:		T_A = –55°C				
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 3.0 A dc; pulsed (see 4.5.1)	h_{FE5}	12		
Subgroup 4						
Pulse response:	3251	Test condition A, except test circuit and pulse requirement in accordance with figure 4				
Pulse delay time			t_d		0.2	µs
Pulse rise time			t_r		1.3	µs
Pulse storage time			t_s		1.2	µs
Pulse fall time			t_f		1.2	µs
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 0.5 A dc; f = 1 kHz	h_{fe}	30	300	
Magnitude of small-signal, short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 0.5 A dc; f = 1 MHz	$\|h_{fe}\|$	4.0	20	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; f = 1 MHz	C_{obo}		500	pF

See footnote at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit	Limits	Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (continuous dc)	3051	T_C = +25°C; t ≥ 1 s; 1 cycle; (see figure 5)				
Test 1		V_{CE} = 15 V dc; I_C = 10 A dc				
Test 2		V_{CE} = 40 V dc; I_C = 3.75 A dc				
Test 3						
2N3715		V_{CE} = 55 V dc; I_C = 0.9 A dc				
2N3716		V_{CE} = 65 V dc; I_C = 0.9 A dc				
Safe operating area (switching) (clamped inductive)	3053	T_A = +25°C; I_C = 10 A dc; V_{CC} = 15 V dc; (see figures 6 and 7)				
2N3715		Clamp voltage = 60 V dc				
2N3716		Clamp voltage = 80 V dc				
Test 1		t_p approximately 1.0 ms (vary to obtain I_C); R_{BB1} = 5Ω; V_{BB1} = 8 V dc; R_{BB2} = 4Ω; V_{BB2} = 0; V_{CC} = 15 V dc; I_C = 5 A dc; L = 4 mH; Q ≥ 100 at 1 kHz; .05 ohms, 20 A dc; (Stanford Miller CK-20 or equivalent)				
Test 2		t_p approximately 2.2 ms (vary to obtain I_C); R_{BB1} = 5Ω; V_{BB1} = 8 V dc; R_{BB2} = 4Ω; V_{BB2} = 0; V_{CC} = 15 V dc; I_C = 2.2 A dc; L = 20 mH; Q ≥ 6,000 at 10 kHz; 0.22 ohms, 12.5 A dc; (two Stancor C-2688 in series or equivalent)				
Electrical measurements		See subgroup 2 of this table				
Subgroups 6 and 7						
Not applicable						

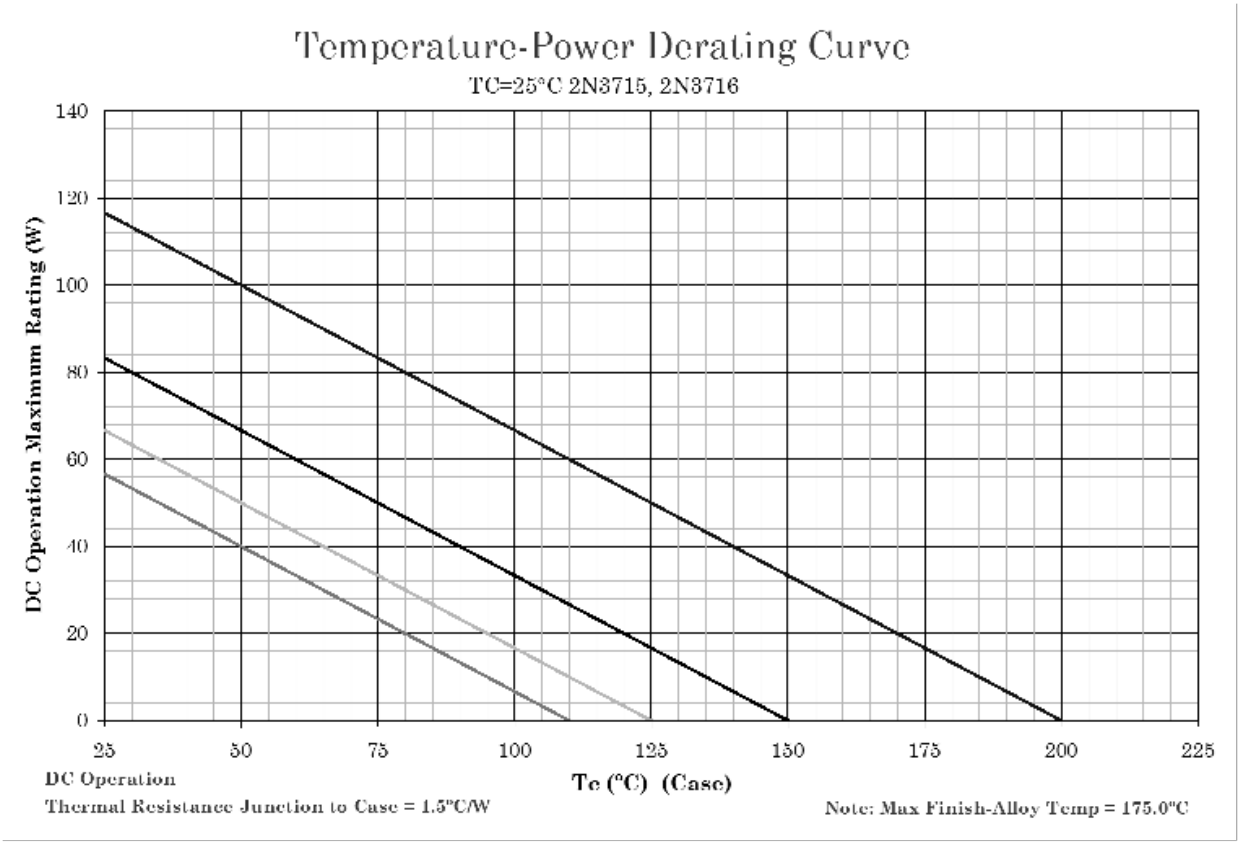
1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3, 4, and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) – for qualification and re-qualification only.

Inspection	MIL-STD-750		Qualification sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C.	
Hermetic seal Fine leak Gross leak	1071		
Electrical measurements		See table I, subgroup 2 and 4.6 herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C; V_{CB} = 80 percent of rated voltage; t = 1,000 hours.	
Electrical measurements		See table I, subgroup 2 and 4.6 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 6			11 devices
Electrostatic discharge sensitivity classification	1020	See MIL-PRF-19500.	
Subgroup 8			45 devices c = 0
Reverse voltage leakage stability	1033	Condition B for devices < 400 V.	

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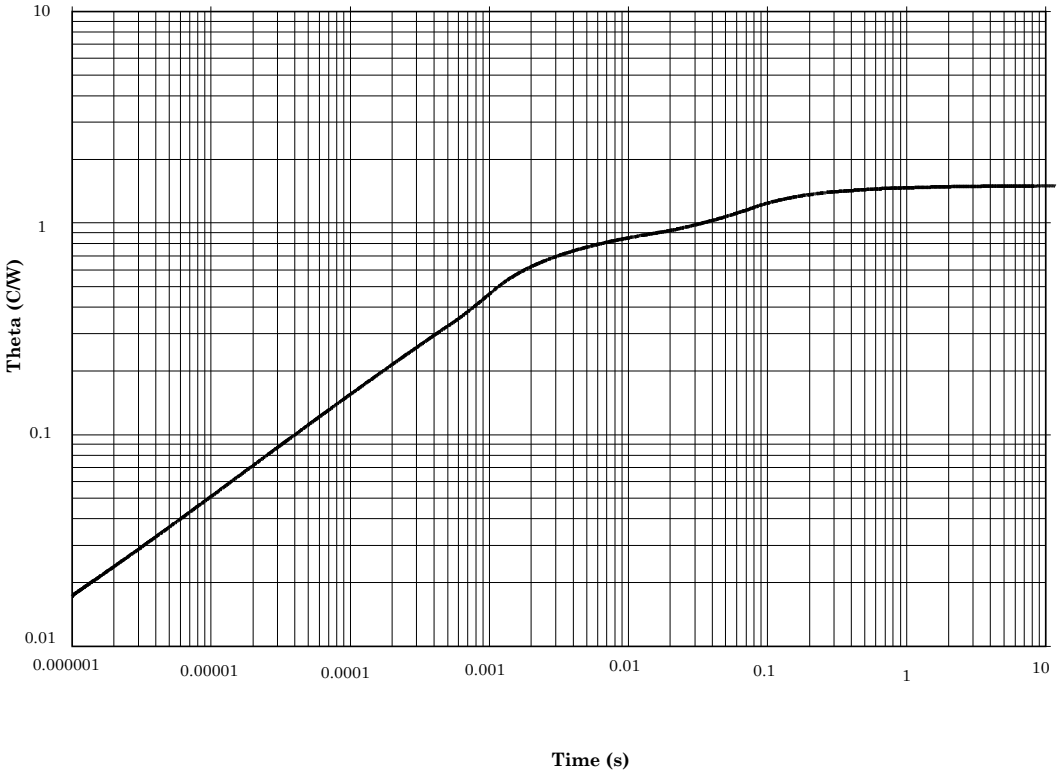
NOTES:

1. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ +200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ +150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves, chosen at T_J ≤ +125°C and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 2. Temperature-power derating graphs for device types 2N3715 and 2N3716.

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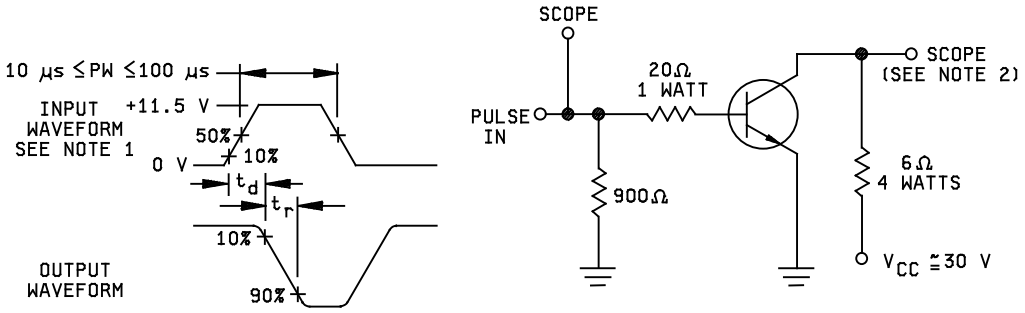
Maximum Thermal Impedance



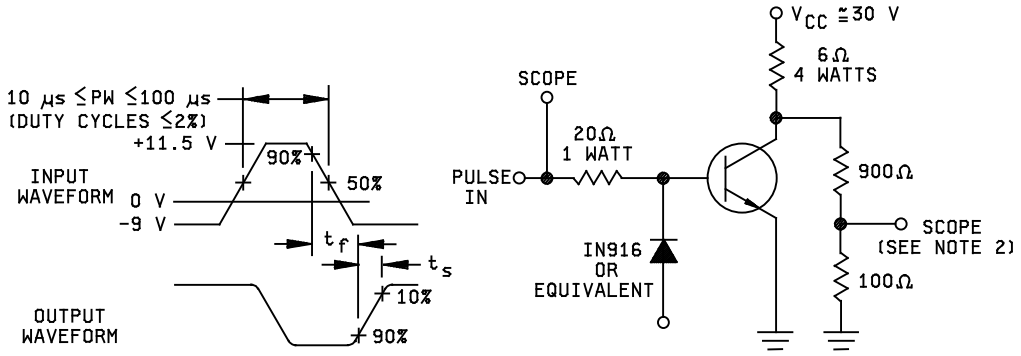
T_C = +25C. Thermal resistance = 1.5°C/W.

FIGURE 3. Transient thermal impedance graph for device types 2N3715 and 2N3716.

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TEST CIRCUIT A



TEST CIRCUIT B

NOTES:

1. The input waveform is supplied by a pulse generator with the following characteristics: $t_r \leq 20.0$ ns, $t_f \leq$ 1iron mamµs, 10 µs ≤ PW ≤ 100 µs, $Z_{OUT} = 50\Omega$, duty cycle ≤ 2 percent.
2. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 5$ ns, $Z_{IN} \geq 100k\Omega$, $C_{IN} \leq 12$ pF.
3. Test circuit A for t_d and t_r; test circuit B for t_s and t_f.

FIGURE 4. Pulse response test circuits.

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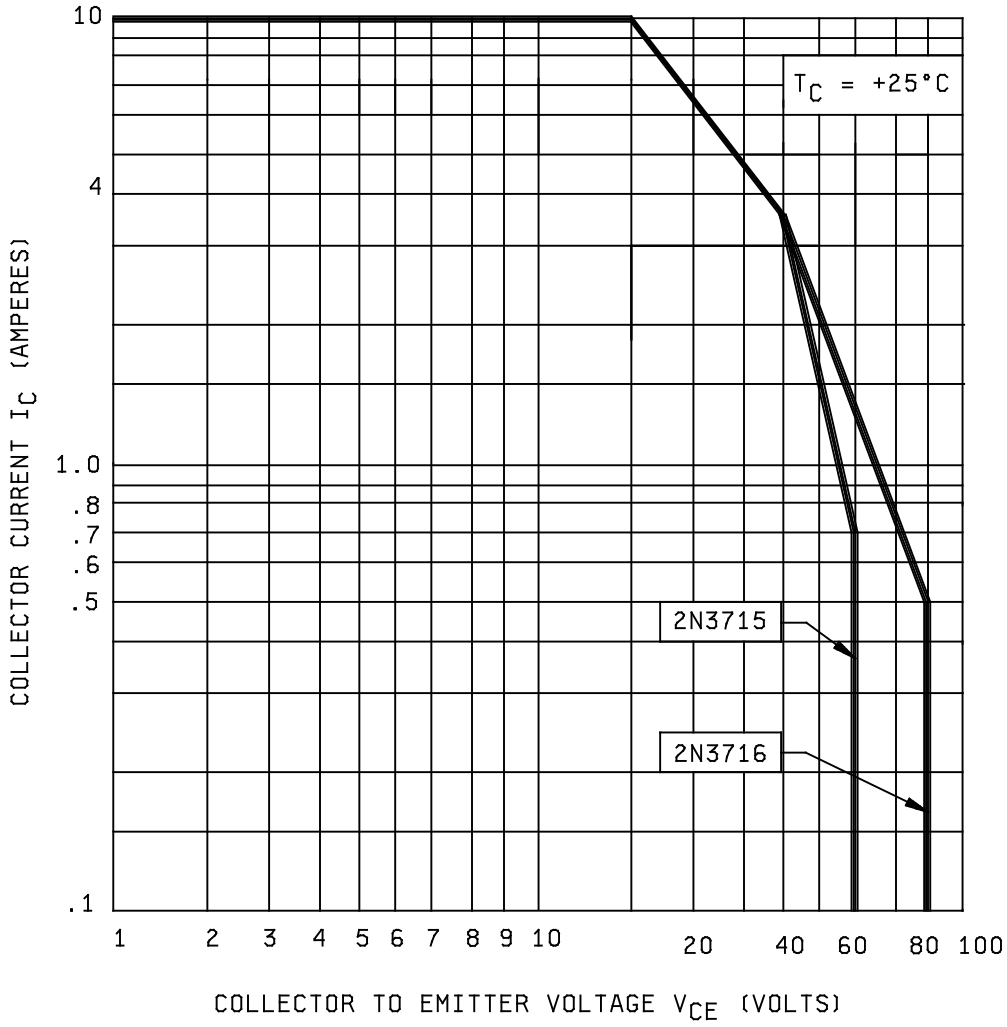


FIGURE 5. Maximum safe operating graph (dc).

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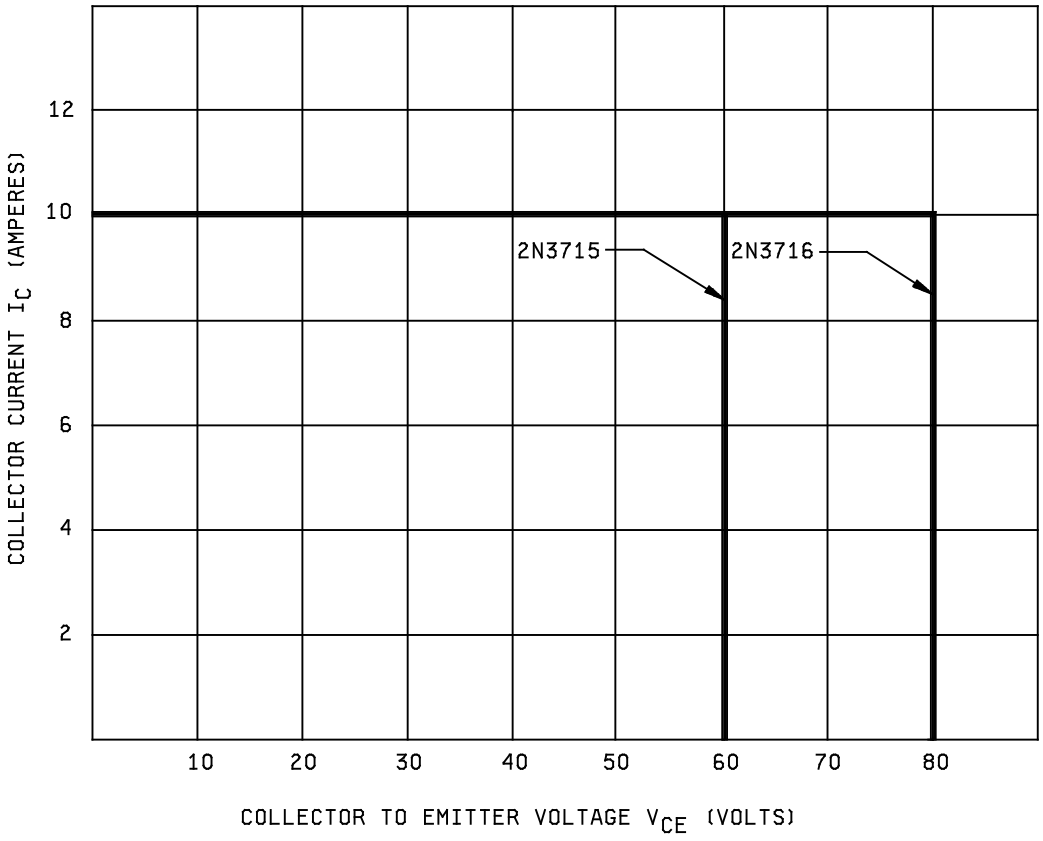
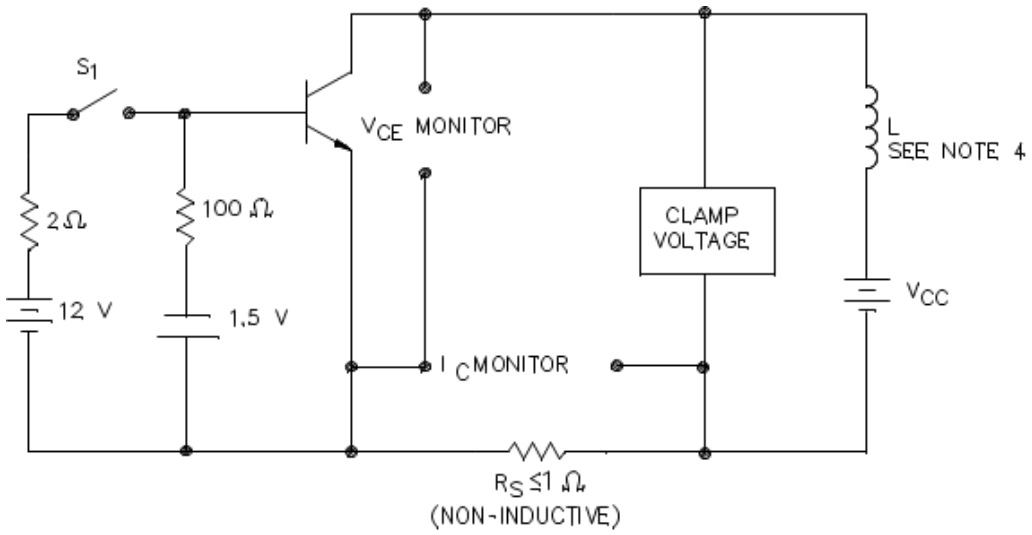


FIGURE 6. Safe operating area for switching between saturation and cutoff (clamped inductive load).

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Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached.
3. Perform specified end-point tests.
4. L = 4.0 mH, 0.05Ω, 20 A, Q ≥ 100 at 1 kHz, (Sanford Miller CK-20 or equivalent).

FIGURE 7. Clamp inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

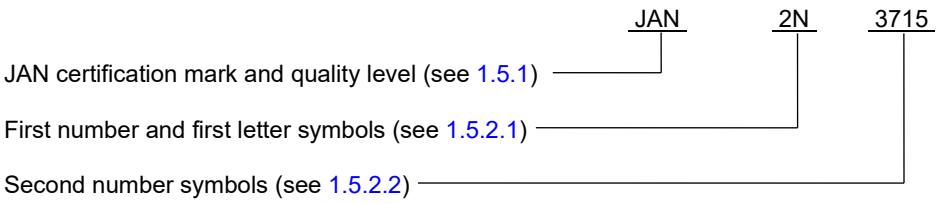
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil. An online listing of products qualified to this specification may be found in the Qualified
* Products Database (QPD) at https://assist.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

JAN quality level	JANTX quality level	JANTXV quality level	JANS quality level
JAN2N3715	JANTX2N3715	JANTXV2N3715	JANS2N3715
JAN2N3716	JANTX2N3716	JANTXV2N3716	JANS2N3716

6.6 Interchangeability information. The device type 2N7368 covered by MIL-PRF-19500/622 is a TO-254 package version of device type 2N3716 covered herein, which is a TO-204AA (formerly TO-3) package version. The military 2N7368 contains the same die as the military 2N3716. The military 2N7368 is preferred over the military 2N3716 whenever interchangeability is not a problem.

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693-1642 or DSN 850-6939.

* 6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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* Navy - SH
Air Force - 85
* NASA - NA
DLA - CC

Review activities:
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* Air Force - 19

Preparing activity:
DLA - CC

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